MILITARY SPECIFICATION

PROJECTILE, 155MM, M718A1/M741A1 TYPES, LOADING, ASSEMBLING AND PACKING OF

This Specification is approved for use within the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembly, and packing for the Projectile, 155MM, M718A1/M741A1 Types.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, listed in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 Ammunition, Standard Quality Assurance Provisions, General Specifications for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSMC N/A

FSC 1320

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MIL-P-48648	Projectile, 155MM, Test, Non-Haz,M718A1/M741A1
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical and Electronic Equipment.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-453	Inspection, Radiographic
MIL-STD-1235	Single and Multi-Level Continuous Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, publication and other Government documents required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER

PRODUCT AND PACKAGEING DRAWINGS

8837839 - Pallet, Projectile, 155MM, HE and GB, Hollow Base Type
9277922 - Mines, AT, Demonstrator, HE
11786215 - Projectile, 155MM, AT, M718A1
11786240 - Projectile, 155MM, AT, M741A1

2.1.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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American Society For Testing and Materials

ASTM Designation D792 - Specific Gravity and Density of Plastics by Displacement, Method A-1.

Code of Federal Regulations - Code of Federal Regulations, Title 49-Transportation parts 100-199.

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103).

2.2 Order of precedene. In the event of a conflict between the text of this specification and the reference cited here, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Projectile. The projectile assembly, mines, and components shall comply with all requirements specified on drawings, all associated drawings, and with all requirements of the applicable specifications.

3.3 S&A assembly. The S&A assembly (9278015) shall be in the full safe condition and have all safety features in the proper position as evidenced by the following:

a. Both centrifugal locks overlapping the slider.

b. G-weight and ball bearing in the safe position.

c. G-weight lock engaged.

d. Set-back pin engaged.

The firing pin of the S&A shall not extend more than .093 inch beyond the housing. There shall be no damage to the MDF; i.e. no cracks in the lead sheathing, or neck down (reduction in diameter) below 75% of the original diameter, anywhere along the length of the MDF.

3.4 Soldering of the flexprint to the electronic assembly terminals. Soldering of the terminals shall comply with MIL-S-45743.

3.5 M-S plate to electronic assembly interface. The M-S plate shall mate with the cover flange of the electronic assembly around the entire circumference. Gaps shall not exceed .025 inch under hand pressure.

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3.6 Explosive output performance

3.6.1 Static functioning of booster (9277827). The booster, when initiated, shall propagate to the main charge, resulting in:

a. High order detonation.

b. Penetration of the M-S plate through the required configuration of steel plates.

c. Spall (particle indentation) on the witness plate. (See 4.5.3)

3.6.2 Static functioning of the main charge (9278018). The main charge, when initiated, shall produce a high order detonation, resulting in:

a. Penetration of the M-S plate through the required configuration of steel plates.

b. Spall (particle indentation) on the witness plate. (Third Plate) (See 4.5.3)

3.7 Ballistic functioning. Representative projectiles, 155MM Test, Non-Haz, M718A1/M741A1 Types (Ref. MIL-P-48648) shall also be included in the ballistic functioninf tests as specified.

3.7.1 Projectile Body. There shall be no premature burst (See 6.5.1) or evidence thereof, and there shall be no separation of metal parts either in the bore of the weapon or in flight, prior to initiation of the nose fuze, in any of the ballistic functioning tests. This requirement shall also apply to those rounds fired at excess pressure in the highest zone of the weapon.

3.7.2 Projectile functioning. The projectile shall function satisfactorily (See 6.5) when subjected to ballistic function tests.

3.7.3 Mine functioning. The mines (See 6.4) shall meet the following requirements after being ejected from the projectile onto the impact area.

3.7.3.1 Functioning time . The mines from the M718A1 projectile shall function within the time specified on drawing 11786215. The mines from the M741A1 projectile shall function within the time specified on drawing 11786240.

3.7.3.2 Functioning after maximum time. The mines from the M718A1 projectile shall not be capable of functioning after the maximum time specified on drawing 11786215. The mines from the M741A1 projectile shall not be capable of functioning after the maximum time specified on drawing 11786240.

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3.7.3.3 Functioning by target vehicle signature. Each mine (See 6.4.b) shall function properly, as evidenced by functioning of the clearing charge when the mine is stimulated by a valid target signature.

3.8 Controls on the use of adhesives. The contractor shall develop a documented operational procedure for utilization of adhesives. This procedure shall provide for quality control of raw materials and processes to be used, details associated with intended use, such as proper handling, dispensing, mixing, curing cycles, pot life, etc., cleaning agents, safety precautions, special ventilation, drying or other processes, and any other information of a similar nature that is considered necessary for the particular adhesive in use. This procedure shall be updated whenever a change is made such that the original assurance of product quality is affected. Submission of this procedure shall be in accordance with Data Item Description DI-T-5372 (Tailored).

3.9 Documentation of plan for safety in handling and processing of munitions. Prior to loading, the contractor shall document a detailed plan of the procedures to be employed during handling of the munition. This documentation shall describe, as a minimum, the procedures to be employed to assure that:

(1) The M718 and M741 electronic assemblies and mines shall remain segregated at all times.

(2) Markings on electronic assemblies, mine assemblies, and projectiles are consistent, and handling of the items controlled to prevent intermixing.

(3) The temperature of the electronic assemblies, mines, or projectile, does not exceed 165°F at any time during processing or storage and the time at 165°F shall not exceed one (1) hour. In no case shall continuous storage of the items be at a temperature above 100°F for a period exceeding 30 days.

(4) High explosive (HE) munition items shall remain segregated from inert munitions at all times, and identification of HE and inert munitions shall be strictly maintained.

Submission of the plan shall be in accordance with data item description DI-T-5372 (Tailored).

3.10 Data requirements. The contractor shall generate data in accordance with the requirements of the data item descriptions cited in paragraph 6.9.

3.11 First article inspection. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein. (See 4.3 and 6.2).

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3.12 Surveillance units. This specification contains provisions for surveillance quantities of Test, Non-Haz Projectiles to be provided for each production lot.

3.13 Shorting bar resistance. There shall be no more than (1) ohm resistance in the shorting bar loop when measured between the output terminal to ground.

3.14 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. All required marking and stamping shall be neat and sharply defined.

Boosters and main charges shall be free of surface defects, (voids, protrusions, inclusions, chips, nicks, etc.). Breakage on the outside edge (corner) of the main charge, not exceeding 1/8 sq. inch in area, and/or 1/8 inch in depth, and not exceeding three (3) per side, is acceptable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample, as designated by the Contracting Officer, for evaluation in

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accordance with the provisions of para. 4.3.2. The first article sample shall consist of the following items in the sample quantities indicated.

Part Description	Drawing	Quantity
Booster	9277827	5
Main Charge	9278018	5
Projectile, 155MM, AT, M718A1	11786215	3
Projectile, 155MM, AT, M718A1 Non Hazardous Test (See 6.4)	11786216	6

4.3.2 Inspections to be performed. The first article parts submitted in accordance with paragraph 4.3.1, will be subjected by the Government to any or all the requirements of the applicable drawings, and to the inspections and tests of Table 1. The projectiles shall be forwarded to a Government Proving Ground designated by the Contracting Officer for the ballistic functioning test.

4.3.3 Rejection. See MIL-A-48078

The first article sample shall be rejected if any assembly or projectile fails to comply with any of the examinations or tests specified in this specification, or any of the requirements of the applicable drawing. In addition, the first article sample shall be rejected if the ballistic test results reveal any one of the defects listed in 4.4.3.9.1.

Defects from the round fired at excess pressure shall be recorded for information, but shall not be included for accept/reject purposes unless a hazardous dud condition is detected. Description of test items and procedures are described in paragraphs 6.4 and 4.5.8 respectively.

In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity. Until the first article quantity is accepted, the contractor is not to proceed with regular production unless otherwise directed by the Contracting Officer.

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TABLE I: FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Projectile, 155MM, AT, M718A1/M741A1				11786215/11786240
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Booster				
	(Dwg. 9277827)				
	Examination for defects	5		3.2	4.4.2.5
	Density	5		3.2	4.5.2
	Explosive performance	5		3.6.1	4.5.3
	Main Charge				
	(Dwg. 9278018)				
	Examination for defects	5		3.2	4.4.2.6
	Density	5		3.2	4.5.2
	Explosive performance	5		3.6.2	4.5.3
	Projectile, 155MM, AT, M718A1				
	(Dwg. 11786215)				
	Examination for defects *	3		3.2	4.4.2.18
	Ballistic tests	3			
	Low zone	1		3.7	4.5.8
	High zone	1		3.7	4.5.8
	Excess pressure	1		3.7	4.5.8

NOTES:

* All applicable paragraphs for mine and projectile assembly apply.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE I: FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Projectile, 155MM,AT, M718A1/M741A1				11786215/11786240
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile, 155MM Test, Non-Haz M718A1 Type (Dwg. 11786216)				
	Examination for defects *	6			MIL-P-48648
	Radiographic Examination	6		3.2	4.5.5
	Ballistic Tests	6	see 4.4.3.10		
	Low zone	3		3.7	4.5.8
	High zone	3		3.7	4.5.8

NOTES:
* All applicable paragraphs for mine and projectile assembly apply.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of projectiles shall contain:

a. Projectile metal parts assemblies of one lot interfix number from one manufacturer.

b. Expulsion charge assemblies from one lot interfix, from one manufacturer.

c. Projectiles of one model designation only.

d. Loaded munitions containing:

(1) S & A assemblies from one interix from one manufacturer.

(2) Explosive from one lot interfix from one manufacturer.

(3) Electronic Assemblies, -1 and -2, from one lot interfix from one manufacturer.

Maximum lot size shall be no more than one (1) month production. (30 calender day period)

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105 Inspection Level II. Equivalent continuous sampling plans in accordance with MIL-STD-1235A may be utilized. The choice of the plan shall be subject to approval from the procuring activity. In any case, inspection for critical defects shall be maintained at 100%.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	S&A Assembly				9278015
					NEXT HIGHER ASSEMBLY
					9278013/9277756
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Safety features improperly positioned		100%	1/	Fluoroscope/4.5.1
Major					
101	MDF Damaged		0.40%	3.3	Visual
102	Output leads improperly positioned or improperly staked		0.40%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES: 1/ See para. 3.3.a, b, c, and d defects only.
2. Data item description DI-R-1721 applies to the major defects. Report required only when defects are found.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Electronic and Primer Assembly (M718A1)				11786214,-1,-2
					NEXT HIGHER ASSEMBLY
					9277756
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Primer missing or improperly installed (improper color showing)		100%	3.2	Visual
102	Bushing missing		100%	3.2	Visual
103	Adhesive present on primer		100%	3.2	Visual
104	Height to top of primer incorrect		0.40%	3.2	Gage/Note 1
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES:
1. Gage shall not damage primer surface or apply unnecessary pressure.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Electronic and Primer Assembly (M741A1)	1 OF 1	11786213-1, 2
			NEXT HIGHER ASSEMBLY: 9278013

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Primer missing or improperly installed (improper color showing)		100%	3.2	Visual
102	Bushing missing		100%	3.2	Visual
103	Adhesive present on primer		100%	3.2	Visual
104	Height to top of primer incorrect		0.40%	3.2	Gage/Note 1
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES:

1. Gage shall not damage primer surface or apply unnecessary pressure.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Battery Check and Insulator Test Point Installation		11786213
			NEXT HIGHER ASSEMBLY
			9278013/9277756

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Battery cell(s) active	1/	100%	3.2	See Dwg. 9317592
102	Test points not resealed		100%	3.2	Visual
103	Test point insulator missing		100%	3.2	Visual
104	Resistance of shorting bar, improper		100%	3.13	4.5.10
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES: 1/ Data Item Description DI-R-1721 applies for this defect. Submission of the report shall be whenever defects are found.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Booster				9277827
					NEXT HIGHER ASSEMBLY
					9278013/9277756
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter incorrect	See 4.4.3.1		3.2	Gage
102	Inside diameter incorrect	See 4.4.3.1		3.2	Gage
103	True position ID to OD incorrect	See 4.4.3.1		3.2	Gage
104	Length incorrect	See 4.4.3.1		3.2	Gage
105	Density incorrect	See 4.4.3.2		3.2	4.5.2
106	Material incorrect				Certification
107	Cracks or laminations		0.40%	3.2	Visual
108	Output performance (See note 1) not proper	See 4.4.3.3		3.6	4.5.3
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES: 1. Data Item Description DI-R-1721 applies to this test. Supply data each time test is performed.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Main Charge		9278018
			NEXT HIGHER ASSEMBLY
			9278013/9277756

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter incorrect	See 4.4.3.1		3.2	Gage
102	Inside diameter incorrect	See 4.4.3.1		3.2	Gage
103	True position OD to ID incorrect	See 4.4.3.1		3.2	Gage
104	Spherical Radius (360° around) improper (2 places)	See 4.4.3.1		3.2	Gage
105	Length incorrect	See 4.4.3.1		3.2	Gage
106	Thickness incorrect	See 4.4.3.1		3.2	Gage
107	Density incorrect	See 4.4.3.2		3.2	4.5.2
108	Cracks or laminations		0.40%	3.2	Visual
109	Explosive output tests improper (See note 1)	See 4.4.3.3		3.6	4.5.3
110	Material improper				Certification
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES:
1. Data Item Description DI-R-1721 applies to the test. Submit each time test is performed.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.7	Mine, Anti-Tank (during assembly) (M741A1)		9278013
			NEXT HIGHER ASSEMBLY 11786240

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	S&A shield over detonators missing, or damaged	3/	100%	3.2	Visual
2	S&A firing pin in partially or fully armed position (See note 1)	1/	100%	3.3	Gage
3	Incorrect electronic assembly part number for the mine identified as M741A1		100%	3.9	Visual
4	Evidence of explosive on mine body threads (prior to assy. of retaining ring)		100%	3.2	Visual
Major					
101	Flex cable twisted or improperly soldered (Ref. note 2B of dwg. prior to potting)		0.40%	3.2/3.4	Visual
102	S&A firing pin tip not visible above S&A housing, or evidence of tip damage		0.40%	3.2	Visual
103	Damage to MDF		0.40%	3.2	Visual

NOTES:

1/ Firing pin extends more than .093 inch beyond housing. Gaging technique shall not touch firing pin tip if below the maximum extension.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.7	Mine, Anti-Tank (during assembly) (M741A1)		9278013
			NEXT HIGHER ASSEMBLY 11786240

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
104	Detonator resistance improper (After soldering of electronic assy)	2/	100%	3.2	Ohmmeter
105	Potting missing or improperly applied to terminals		0.40%	3.2	Visual
106	Insulator missing over terminals		100%	3.2	Visual
107	Clearing charge assy/cup with collar improperly assembled (Ref note 2B of dwg)		0.40%	3.2	Visual
108	Plate not seated on cover flange (360° around)		0.40%	3.5	Visual
109	O-ring missing (ref note 2C of dwg)		100%	3.5	Visual
110	Adhesives improperly applied (See notes 2C and 6 of dwg)	See 4.4.3.4		3.2	Visual
111	Cracking of main charge (After pre-load)	See 4.4.3.5		3.2	Visual
112	Plate missing (See note 2E of dwg)		0.40%	3.2	Visual
113	Impact lens or cover missing (See note 2F of dwg)		0.40%	3.2	Visual
114	Retaining ring height improper		0.40%	3.2	Visual/Gage
115	Mine pre-load/retaining ring torque improper	See 4.4.3.6		3.2	4.5.7
116	Radiographic examination				
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES: 2/ Less than one (1) milliampere current to be applied to detonators.
3/ Data item description DI-R-1721 applies to the critical defects, supply report whenever defects are found.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Mine, Anti-Tank (during assembly) (M718A1)	1 OF 2	9277756
			NEXT HIGHER ASSEMBLY 11786215

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	S&A shield missing, or damaged over detonators	3/	100%	3.2	Visual
2	S&A firing pin in partially or fully armed position	1/	100%	3.3	Gage
3	Incorrect electronic assembly part number		100%	3.9	Visual
4	Evidence of explosive on mine body threads (prior to assy. of retaining ring)		100%	3.2	Visual
Major					
101	Flex cable twisted or improperly soldered (Ref. note 2B of dwg. prior to potting)		0.40%	3.2/3.4	Visual
102	S&A firing pin tip not visible above S&A housing, or evidence of tip damage		0.40%	3.2	Visual
103	Damage to MDF		0.40%	3.3	Visual

NOTES: 1. Firing pin extending .093 inch beyond housing. Gaging technique shall not touch firing pin tip if below the maximum extension.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9277756
4.4.2.8	Mine, Anti-Tank (during assembly) (M718A1)				NEXT HIGHER ASSEMBLY 11786215
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
104	Detonator resistance improper (After soldering to electronic assy)	2/	100%	3.2	Ohmmeter
105	Potting missing or improperly applied to terminals		0.40%	3.2	Visual
106	Insulator missing over terminals		100%	3.2	Visual
107	Clearing charge assy/cup with collar improperly assembled (Ref note 2B of dwg)		0.40%	3.2	Visual
108	Plate not seated on cover flange (360° around		100%	3.2	Visual
109	O-ring missing (ref note 2C of dwg)		100%	3.2	Visual
110	Adhesives improperly applied (See notes 2C and 6 of dwg)	See 4.4.3.4		3.2	Visual
111	Cracking of main charge (After pre-load)	See 4.4.3.5		3.2	Visual
112	Plate missing (See note 2E of dwg)		0.40%	3.2	Visual
113	Impact lens or cover missing (See note 2F of dwg)		0.40%	3.2	Visual
114	Retaining ring height improper		0.40%	3.2	Visual
115	Mine pre-load/retaining ring torque improper	See 4.4.3.6		3.2	4.5.4
116	Radiographic examination				4.5.7
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES: 2/ Less than one (1) milliampere. Current to be applied to detonators.
3/ Data item description DI-R-1721 applies to the critical defects and major 101; submit whenever defects are found.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Projectile, 155MM, M718A1/M741A1 Metal Parts (Prior to Loading Mines)	1 OF 1	10542902
			NEXT HIGHER ASSEMBLY: 11786215/11786240

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Rust or foreign matter on projectile interior walls, threads, or base end		0.40%	3.2	Visual
102	Threads damaged on projectile ogive or base ends		0.40%	3.2	Visual
103	Base threads damaged		0.40%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Projectile, 155MM, AT, M741A1 (Prior to Inserting Mines into Projectile)	1 OF 1	11786240
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Incorrect (M718A1) or unidentified mine(s) being assembled into projectile designated M741A1 or projectile not identified	1/	100%	3.9	Visual
Major					
101	Key or any spherical pad missing from any mine		100%	3.2	Visual
102	Pusher plate missing		100%	3.2	Visual
103	Improper number of -1 and -2 mines		100%	3.2	Visual
104	Correleratan of Mine color code and Explosive loading of Mine		100%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES: 1/ Data item description DI-R-1721 applies for the critical defects.Submission of report shall be whenever defects are found. Immediate notification of technical agency required for critical defects.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Projectile, 155MM, AT, M741A1 (After Inserting Mines & Prior to assembling base				11786240
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Mine stack preload improper	See 4.4.3.7		3.2	4.5.4
102	Shim thickness improper	See 4.4.3.7		3.2	Gage
103	Evidence of grease on base or projectile threads		0.40%	3.2	Visual
104	Sealant missing from base threads		0.40%	3.2	Visual
105	Base O-ring missing		0.40%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.12	Projectile, 155MM, AT, M741A1 (Prior to assembling lifting plug)	1	1	11786240
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Base improperly torqued	See 4.4.3.8		3.2	Visual/4.5.4
102	Gap between body and base improper		0.40%	3.2	Gage
103	Expulsion charge assembly missing, damaged, or cup improperly staked		100%	3.2	Visual/Manual
104	Projectile fails air pressure retention test	1/	100%	3.2	4.5.5
105	Fuze thread in ogive damaged, incomplete or contains foreign matter		100%	3.2	Visual
106	Projectile zone weight incorrect		100%	3.2	Scale
107	Zone stenciling or zone punch mark incorrect or inadequate		100%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES: 1/ 100% test. Verify production test or sample following 100% production test to MIL-STD-105 level II. Reject lot if one (1) defective found. Use approved equipment.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.13	Projectile, 155MM, Test, M741A1 Type				11786241
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Nomenclature on body, letter in triangle and stamped not consistent	1/	100%	3.9	Visual
2	"M" missing from body	1/	100%	3.2	Visual
3	Radiographic examination of Ballistics Mines and projectiles	2/			
Major					
101	Grease missing from lifting plug threads (prior to installing)		0.40%	3.2	Visual
102	Lifting plug missing or improper		0.40%	3.2	Visual
103	Lifting plug torque improper		0.40%	3.2	Torque wrench
104	Lifting plug or gasket missing or damaged		100%	3.2	Visual
105	Max. diameter forward of grommet improper (Applicable after retouching paint)		100%	3.2	Gage
106	Stamped letter "A" missing from base		100%	3.2	Visual
107	Rotating band damaged (Applicable if grommet has been removed		0.40%	3.2	Visual
108	Obrurator missing or damaged (Applicable if grommet has been removed)		0.40%	3.2	Visual

NOTES: 1/ Data item description DI-R-1721 applies for all critical defects. Submission of the report shall be whenever defects are found. Immendiate notification of technical agency for critical defects required. 2/ Applicable to Proving Ground prior to Ballistic Test.

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PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.13	Projectile, 155MM, AT, M741A1				11786240
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Protective coating with a total of bare areas in excess of 1/4 square inch		0.65%	3.2	Visual
202	Marking missing or unidentifiable (Other than critical defect 1 above)		0.65%	3.2	Visual
203	Poor workmanship		0.65%	3.14	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Projectile, 155MM, AT, M741A1 Ballistic test				11786215
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Projectile in flight detonation	1/	4.4.3.9.1	3.7.1	4.5.8/Visual
2	Late mine function after SD time (Hazardous dud)	1/	4.4.3.9.1	3.7.3.2	4.5.8
Major					
101	Improper projectile flight/function	1/	4.4.3.9.1	3.7.2	4.5.8/6.5.2
102	Mine performance improper	1/	4.4.3.9.1	3.7.3	4.5.8.3
Minor	None defined				

NOTES:

1/ See paragraph 4.4.3.9.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Projectile, 155MM, AT, M718A1(Prior to inserting mines into projectile)	1 OF 1	11786215
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Incorrect(M741A1) or unidentified mines assembled into projectile designated M718A1 or projectile not identified		100%	3.2	Visual
102	Key or any spherical pad missing from any mine	1/	100%	3.2	Visual
103	Pusher plate missing		100%	3.2	Visual
104	Proper number of -1 and -2 mines		100%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES: 1/ Data item description DI-R-1721 applies for Major 101. Submit report whenever defects are found. Immediate notification of technical agency for critical defects required.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.16	Projectile, 155MM, AT, M718A1(After Inserting Mines & Prior to assembling base)				11786215
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Mine stack preload improper	See 4.4.3.7		3.2	4.5.4
102	Shim thickness improper	See 4.4.3.7		3.2	Gage
103	Evidence of grease on body or base threads		0.40%	3.2	Visual
104	Adhesive missing from base threads		0.40%	3.2	Visual
105	Base O-ring missing		0.40%	3.2	Visual
Minor	Poor workmanship		0.65%	3.14	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.17	Projectile, 155MM, AT, M741A1(Prior to assembling lifting plug)				11786215
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Base improperly torqued	See 4.4.3.	8	3.2	Visual/4.5.4
102	Gap between body and base improper		100%	3.2	Gage
103	Expulsion charge assembly missing, damaged, or cup improperly staked		100%	3.2	Visual/Manual
104	Projectile fails air pressure retention test	1/	100%	3.2	4.5.5
105	Fuze thread in ogive damaged, incomplete or containing foreign matter.		100%	3.2	Visual
106	Projectile zone weight incorrect		100%	3.2	Scale
107	Zone stenciling and/or zone punch marks incorrect or inadequate		100%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES: 1/ 100% test. Verify production test or sample following 100% production test to MIL-STD-105 level II. Reject lot if one (1) defective found. Use approved equipment.

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MIL-P48649(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.18	Projectile, 155MM, AT, M718A1				11786215
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Nomenclature on body, and letter in triangle not consistent		100%	3.9	Visual/Note 1
2	"M" missing from body		100%	3.2	Visual/Note 1
3	Radiographic examination of Ballistic Mines and Projectiles	2/	100%	3.2	4.5.7
Major					
101	Grease missing from lifting plug threads (prior to installing)		0.40%	3.2	Visual
102	Lifting plug improper or missing		0.40%	3.2	Visual
103	Lifting plug torque improper		0.40%	3.2	Torque wrench
104	Lifting plug gasket missing or damaged		100%	3.2	Visual
105	Max. diameter forward of grommet improper (Applicable after retouching paint)		100%	3.2	Gage
106	Stamped letter "A" missing from base		0.40%	3.2	Visual
107	Rotating band damaged (Applicable if grommet has been removed		0.40%	3.2	Visual
108	Obturator missing or damaged (Applicable if grommet has been removed)		0.40%	3.2	Visual

NOTES: 1/ Data item description DI-R-1721 applies for all critical defects. Submission of the report shall be whenever defects are found. Immediate notification of technical agency for critical defects required. 2/ Applicable to proving ground prior to ballistic test.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.18	Projectile, 155MM, AT, M718A1				11786215
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Protective coating with a total of bare areas in excess of 1/4 square inch		0.65%	3.2	Visual
202	Marking missing or unidentifiable (Other than critical defect 1 above)		0.65%	3.2	Visual
203	Poor workmanship		0.65%	3.14	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.19	Projectile, 155MM, AT, M718A1 Ballistic Test		11786215
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Projectile in flight detonation	4.4.3.9	4.4.3.9.1	3.7.1	4.5.8/Visual
2	Late mine function after SD time (Hazardous dud)	4.4.3.9	4.4.3.9.1 4.4.3.9.2	3.7.3.2	4.5.8/6.6
Major					
101	Improper projectile flight/function	4.4.3.9	4.4.3.9.1	3.7.2	4.5.8/6.5.2
102	Mine performance	4.4.3.9	4.4.3.9.1	3.7.3	4.5.8.3
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48649(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.20	Pallet (loaded)	1	1	8837839
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cover support missing (P/N 925390)	1/	100%	3.2	Visual (8 places)
102	Projectile grommet assembly missing damaged, or improperly assembled		0.40%	3.2	Visual
103	Board broken or split		0.40%	3.2	Visual
104	Projectile(s) of different zones on the same pallet		0.40%	3.2	Visual
105	DOT symbol missing or unidentifiable		0.40%	3.2	Visual
Minor					
201	Strap width incorrect		0.65%	3.2	Visual/Gage
202	Strapping improperly assembled		0.65%	3.2	Visual
203	Marking missing, incorrect or unidentifiable		0.65%	3.2	Visual
204	Poor workmanship		0.65%	3.14	Visual

NOTES:
1/ Cover Support is required with Fusible Lifting Plug P/N 9215390. Universal Lifting Plug Assembly P/N 9345325 does not require Cover Support.

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.3.1 Dimensional control of main charges and boosters (See dwg 9277827 and 9278018). The dimensions specified in paragraphs 4.4.2.5 and 4.4.2.6 for the boosters and main charges shall be checked by randomly selecting three (3) samples from the quantity produced by each press during a period not to exceed four (4) hours. The samples shall be allowed to cool and become dimensionally stable prior to inspecting. The quantity represented by the samples shall be identified and not used in production until the dimensional inspections are completed. If any one (1) sample fails to comply with the dimensional requirements, the quantity of parts represented by the sample shall be subjected to 100% inspection. The contractor and technical agency shall determine the amount of time required for cooling and supply this information to the resident Government inspector. The above inspection shall also be performed each time a new lot of explosive material is used.

4.4.3.2 Density of the main charge and booster (see dwg. 9277827 and 9278018. The density of the main charges (each press) and boosters (1 press) shall be checked by randomly selecting one (1) sample from the quantity produced by each press during each four (4) hour period. The boosters and main charges shall be allowed to cool, become dimensionally stable, and meet dimensional requirements prior to performing the density measurement. The quantity represented by the sample shall be identified and not used in production until the density measurement is made and the part accepted. If the sample fails to comply with the density requirements of the applicable drawing, the quantity of parts represented by the sample shall be subjected to 100% inspection. The Contractor and technical agency shall determine the amount of time required for cooling and supply this information to the resident Government inspectors. The above test shall also be performed each time a new lot of explosive material is used.

4.4.3.3 Explosive performance. A sample of each lot of explosive used to manufacture main charges and boosters shall be tested at the start of each new explosive lot usage. One (1) main charge shall be randomly selected and inspected for the defects listed in 4.4.2.6. The density of the samples shall be determined (see 4.5.2) and if found to be within acceptable limits, the sample shall be set aside for explosive performance tests. The main charges shall be tested in accordance with 4.5.3, using booster from a previously accepted booster lot.

Similarly, one (1) booster shall be randomly selected and inspected for the defects listed in paragraph 4.4.2.5. The density of the sample shall be determined (see 4.5.2) and if found to be within acceptable

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limits, the sample shall be set aside for explosive performance tests. The booster shall be tested in accordance with 4.5.3, using a main charge from a previously accepted main charge lot.

Failure of the explosive performance sample to comply with the requirements of 3.6 shall result in rejection of the lot of explosive material represented by the sample. Results of these tests shall be supplied in accordance with para 6.8.

4.4.3.4 Adhesive application. At the start of each days production, and after each four (4) hours during production, and at the end of each shift, a verification of proper adhesive application shall be made, following the procedure generated by the contractor (see 3.8). The procedure shall assure that the proper amount of adhesive has been and is being dispensed and properly applied. The method shall assure that adhesive is not migrating to the S&A assembly. Additionally, all other process controls delineated in para. 3.8 shall be verified, at least once each shift. Evidence of inadequate adhesive to the S&A during any of the verification checks shall result in rejection of that quantity of mines assembled since the last acceptable check. If the adhesive application does not comply with all requirements at the start of production the Contractor shall not proceed with production until corrective action has been taken and all requirements met.

4.4.3.5 Cracking of the main charge. For each new lot of plates (9277742), main charges (9278018) or electronic assemblies (9342551 & 9342552) used in production, a sample mine shall be assembled without applying adhesive. The mine shall be pre-loaded and torqued in accordance with the mine drawing requirements. After application of the preload and torque, the retaining ring, cover, impact lens and plate shall be removed, and the main charge examined for cracks. If there is any evidence of cracking of the main charge, production shall be halted, and the Contracting Officer and Technical agency notified. If there is no evidence of main charge cracking, the unit may be returned to the lot, after application of the adhesive, for further assembly.

4.4.3.6 Mine preload and retaining ring torque. At the start of each shift, and after each four (4) hours of production, and at the end of each shift, the mine preload and retaining ring torque shall be verified. The Contractor shall supply for approval, a plan detailing the method to be used to verify these forces (see 6.8). The contractor shall not proceed with production until verification is achieved. If the control checks indicate that the proper forces are not being applied, the mines assembled since the last acceptable checks shall be subjected to 100% inspection. The contractor may request a change in inspection frequency provided he presents objective evidence to the contracting officer to substantiate the request.

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4.4.3.7 Mine stack preload and shim thickness measurement. At the start of each shift, and after each four (4) hours of production, and at the end of each shift, the mine stack preload force and shim thickness measurements shall be verified. The Contractor shall supply, for approval, a plan detailing the methods to be used to verify these requirements (see 6.8). The contractor shall not proceed with production until verification of proper operation has been achieved. Additionally, the contractor shall verify that the proper shim thickness is being added to the projectiles. If the verification check indicates that the mine stack preload or shim thickness is incorrect, the quantity of projectiles assembled since the last acceptable check shall be subjected to 100% inspection.

4.4.3.8 Base torque. At the start of each shift, and each four (4) hours thereafter, and at the end of each shift, the base torque shall be verified. The contractor shall supply for approval, a plan detailing the method to be used to verify the base torque (see 6.8). The contractor shall not proceed with production until the proper torque has been demonstrated. If the verification check indicates that the base torque is incorrect, the quantity of projectiles assembled since the last acceptable check shall be subject to 100% inspection.

4.4.3.9 Ballistic test. The quantity and type of projectile from each lot to be subjected to ballistic tests shall be as specified below. Lots of projectiles shall conform to the lotting provisions of 4.4.1 but in no case shall exceed one (1) months production. HE Projectiles shall be randomly selected from the lot. Non-Hazardous Projectiles (para. 6.4.b) shall comply with the lotting requirements of MIL-P-48648.

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TEST CONFIGURATION/TEST ZONE (SEE 6.4)

TOTAL TEST QTY PER SYSTEM	HE TYPE QTY	HE TYPE ZONE	NON HAZ TEST QTY	NON HAZ TEST ZONE	TEST QTY OF MINES/ TEST PERFORMED
8-M718A1 Proj	1	3	3	3	9/Self Dest. 18/ Sensor
	1	7	3	7	9/Self Dest. 18/ Sensor
8-M741A1 Proj	1	3	3	3	9/Self Dest. 18/Sensor
	1	7	3	7	9/Self Dest 18/Sensor

4.4.3.9.1 Rejection. The lot of projectiles represented by the ballistic sample shall be rejected if the ballistic test results reveal any one of the following defects:

a. Any premature burst occurs (see 6.5.1).

b. Any separation of metal parts occurs prior to initiation of the nose fuze.

c. Any detonation of the projectile cargo occurs prior to initiation of the nose fuze.

d. Any projectile fails to function satisfactorily (see 6.5).

e. A total of five (5) or more of the seventy-two (72) HE and Non Hazardous Test Type mines tested fail to function at any time in any mode.)NOTE: Failure of the PA506 Delay Detonator to function is considered as a dud mine.)

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f. Two (2) or more mine functions are capable of functioning after the maximum self-destruct time has elapsed. (See 4.4.3.9.2)

g. A total of eight (8) or more of the seventy-two (72) mines tested function prior to minimum self-destruct time or functioned prior to sensor performance test

h. A total of five (5) or more of the thirty-six (36) Non Hazardous Test mines fail to function when subjected to the target vehicle signature.

4.4.3.9.2 Lot referral. If one (1) mine is determined to have functioned after the maximum self-destruct time has elapsed, and all other acceptance criteria has been met, lot deposition shall be referred to AMCCOM ATTN: AMSMC-QAL-N, Rock Island, ILL. Acceptance or rejection will be based on a statistical determination of system compliance with the requirement for hazardous duds based on accumulated LAT data.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit inspection equipment designs for the approval in accordance with the terms of the contract see section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures

4.5.1 S&A fluoroscopic examination. S&A assemblies shall be examined for presence and position of the safety features (See para. 3.3(a) thru (d). The quality of the fluoroscope shall be such to permit ready identification of the presence and location of the parts. Rotation of the part may be necessary and is permitted. Any assembly having a safety feature missing or out of position shall be immediately removed from the lot and marked or tagged to prevent the assembly from entering subsequent operations. Data item description DI-R-1721 applies to this paragraph. Submission shall be weekly. Immediately notify technical agency of defective part.

4.5.2 Density of the main charge and boosters. Density of the main charge (9278018) and booster (9277827) shall be performed in accordance with the procedure specified in ASTM Designation D792 - "Specific Gravity and Density of Plastics by Displacement", Method A-1. The contractor may use alternate procedures than that specified herein, subject to the approval of the Government.

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4.5.3 Explosive output performance. The main charges or boosters to be tested (See 4.4.3.3) shall be assembled in accordance with drawing 9277922. The mine assembly shall be functioned under the plates configured in fig. 1 below following approved safety procedures.

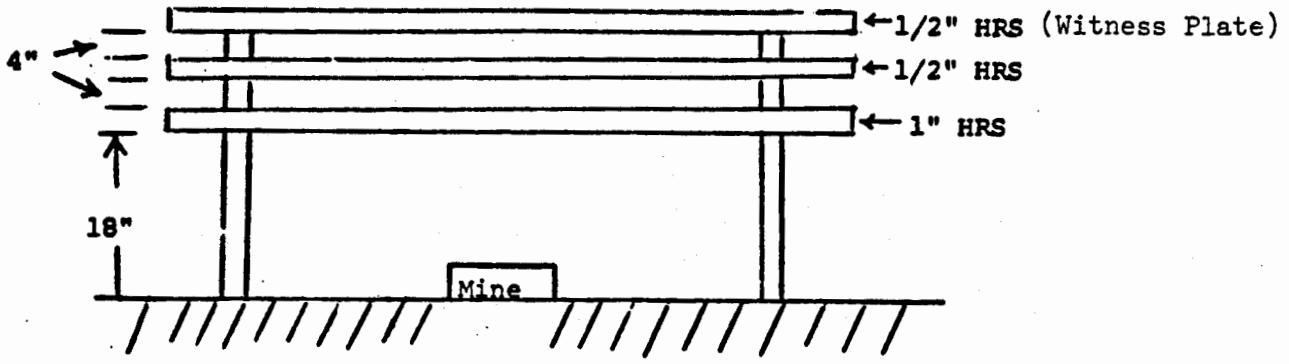


FIG. 1

4.5.4 Preload and torque measurements. Preload and torque measurements shall be performed in accordance with the plans submitted in accordance with paragraphs 4.4.3.6, 4.4.3.7, 4.4.3.8, and 4.4.3.9 as applicable.

4.5.5 Air pressure test of projectile. The assembled projectile, with lifting plug, shall be placed in a chamber and a pre-determined volume of air shall be delivered to produce fifteen plus one (15 + 1) pounds per square inch (psi) pressure between the outside of the projectile and the interior of the chamber. After initial stabilization of the pressure the projectile shall withstand the external pressure (15 + 1 psi) for fifteen (15) seconds minimum without exhibiting more than a 0.1 psi drop in pressure. Any projectile failing this test shall be rejected, suitably marked, and removed from the lot. All equipment used in this test shall be capable of performing the test accurately and consistently. Equipment shall be calibrated at intervals to be approved by the technical agency. If it is found that the gage is out of calibration in such a direction that the test pressure was actually lower than that specified on the drawing, projectiles tested since the last satisfactory calibration shall be retested after the error has been corrected. All test and calibration instructions, equipment and procedures shall be approved in accordance with 4.4.4.

4.5.6 Projectile weight determination. The projectile, without lifting plug, gasket, and grommet assembly shall be weighed, using equipment of suitable range and accuracy. The zone weight shall be marked in the specified location on the projectile and this marking shall be verified as being proper for the weight zone indicated by the equipment.

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4.5.7 Radiographic examination of projectiles for ballistic test. Radiographs shall be taken of the test projectiles selected for ballistic test. The projectiles shall be radiographed at the proving ground before test firing. The radiographs of the projectiles shall be taken with the beam focused perpendicular to the center axis 90° from the keyway mark. The location of the g'weight, ball, and slider shall be visible in the radiographs. The non-hazardous projectiles must show evidence of the lead shot in the booster simulant and main charge simulant of each mine and dummy center lead of each S&A. The beam may be moved to obtain a picture of all the mines in the projectile. The radiographs may be retained for inspection upon request or forward to ARDC, ATTN: DRSMC-QAT-M(D). Any indication of a missing or out-of-line g-weight or ball in any S&A or missing lead shot in the booster or main charge simulants or dummy center lead shall result in that projectile being clearly marked and removed from test and the Contracting Officer and Technical Agency immediately notified.

NOTE: Radiographic equipment, operations and procedures shall be qualified in accordance with MIl-STD-453. In addition, the photographic density of the film used shall be of quality which will produce a sharp definition of internal structure.

4.5.8 Ballistic functioning

4.5.8.1 Preparation and firing. Ballistic test shall be conducted at a Government proving ground in accordance with approved Acceptance Test Procudures. All ballistic test rounds shall be radiographically examined in accordance with 4.5.7 prior to test. Unless otherwise specified, all LAT projectiles and propellant charges shall be maintained at a temperature of seventy (70) degrees fahrenheit plus or minus ten (10) degrees for a minimum of twenty-four (24) hours prior to firing. Conditioning of the rounds in a temperature chamber will be performed only as needed to achieve the above. The round shall be fitted with a fuze which is standard for the projectile. The projectile fuze setting and the quadrant of elevation of the weapon shall be adjusted according to the firing tables for the system. A photograph shall be taken of the fuze setting for each round prior to firing and each photograph shall be marked for future identification with the round. The sample projectile(s) shall be tested in the specified zone(s) using weapon, equipment, methods, and procedures in accordance with the acceptance test procedures referenced above. The HE rounds shall be fired into the prepared test area at Yuma Proving Ground that has previously been used for M718/M741/HE testing (Mullins Square). The M109A1 weapon shall be used unless otherwise specified by the procuring agency.

4.5.8.2 Projectile.

a. Low zone. The sample projectiles to be subjected to this test shall be fired using the propelling charge for zone 3 in the M109A1 weapon. Observation shall be made for failure to comply with the requirements of 3.7.

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b. High zone. The sample projectiles to be subjected to this test shall be fired using the propelling charge for zone 8 in the M109A1 weapon. Observation shall be made for failure to comply with the requirements of 3.7.

c. Excess pressure. The sample projectiles to be subjected to this test shall be fired using the propelling charge adjusted to produce the requqired pressure. (See 6.5.4) Observation shall be made for failure to comply with the requirements of 3.7.1. Burst height and cargo functioning are not considered a part of this test.

4.5.8.3 Munition. After ejection and impact, the munition shall be evaluated for compliance with the requirements of 3.7.3 as follows:

a. HE mines (See 6.4). Recording equipment shall be used to determine the time of functioning of the HE mines. It is recommended that a delay of one (1) hour be observed when firing more than one (1) round of the same type (i.e. two (2) M718's). Monitoring shall continue for a minimum of four (4) hours beyond the maximum self-destruct time specified on the drawing if it is determined that all mines have not detonated. HE mines shall only be monitored (from a safe distance) for self-destruct functioning. No attempt shall be made to retrieve the mines. The times of mine functioning shall be recorded. Timing shall begin from the time the mines are ejected from the round.

b. Test Non Haz Type mines (See 6.4). Test Non Haz Type mines shall be located on the impact field and observed for self-destruct functioning as soon as possible after impact. The concave side cover shall be carefully examined (minimum tilting possible) for bulging and/or expulsion from the mine. Mines found with bulging covers at this time shall be recorded as early functioning.

(1) Self-destruct functioning. Mines selected for self-destruct functioning shall have a small stone or other convenient object placed on the top surface of the mines that have not functioned as an indicator for subsequent mine functioning (mine functioning should cause sufficient disturbance to dislodge the stone). The mines shall then be monitored for self-destruct functioning. Time of functioning shall be recorded. Timing begins from the time the mines eject from the round.

If any mine fails to function by maximum self-destruct time, an attempt shall be made to function the mine by using a target simulator (to be supplied by the technical agency). Failure of the mine to function, reflects a non-hazardous dud. Functioning after the maximum self-destruct time, reflects a hazardous dud.

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NOTE: Mines with AD switches may function when disturbed. This functioning shall be reported and is not considered a failure of the item.

CAUTION: Covers may fly off and the mine may jump approximately six (6) inches at time of functioning. All handling shall be done with the concave cover facing away from all test personnel preferrably from behind a plexiglass barrier.

(2) Sensor performance. Mines selected for sensor performance test that have not functioned early shall be functioned by the target simulator test box supplied by the technical agency. If any mine fails to function by the simulator, as evidence by audible sound of the detonators functioning, it shall be monitored for self-destruct functioning as specified in (1) above. If a mine fails to function by the maximum self-destruct time, a second attempt will be made to function the mine by the simulator. Failure of the mine to function reflects a non-hazardous dud functioning after SD time out reflects a hazardous dud.

(3) Verification of MDF functioning. All non-hazardous mines shall be disassembled following approved procedures and examinations made for functioning of both detonators and the MDF assembly. The proving ground shall forward a copy of the test procedure and disassembly procedure to the technical agency specified in 6.6 for review and approval.

(4) Dud Monitoring. Unless otherwise directed by the Technical Agency, all non-haz type mines that fail to function and are concluded to be duds, shall be removed to a storage area, identified by the projectile lot number, and held for a period of sixty (60) days. At the end of this period, the mines shall be examined for any evidence of functioning. Results of this examination shall be reported to the Technical Agency specified in 6.6.

4.5.8.4 Records of ballistic tests. A record shall be made for each round fired and shall include an accounting for the defects listed in 4.3.2 or 4.4.3.11 as applicable. In addition, the record shall include nose fuze duds or malfunctions, expulsion height, actual munition functioning time including early functions, and any other pertinent data. A copy of the ballistic test results for each projectile lot shall be forwarded to the address specified in 6.6.

4.5.8.5 Nose fuze malfunctions. Nose fuze malfunctions shall not be considered as cause for rejection of a projectile lot. When a fuze malfunction occurs, the proving ground shall notify the technical agency and request a determination as to the need for replacement projectile.

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4.5.9 Surveillance units. The Test, Non-Haz Projectiles designated for surveillance are to be stored at the load plant until disposition is made. The projectiles are to be identified and serialized in a manner to provide traceability to month and year of manufacture (see MIL-P-48648). The following provides the number of Test, Non-Haz M718 or M741 Type Projectiles that are required for the respective production lot quantities.

Lot Quantity	No. Proj. Test, Non-Haz
Less than 500	6
500 to 1500	8
Greater than 1500	16

4.5.10 Shorting bar resistance. The resistance of each shorting bar circuit shall be measured between the test points E3 and TP9, using an approved ohmmeter. Observation shall be made for compliance with the requirements of 3.13.

5. PACKAGING.

5.1 Packing (Level A). The projectiles shall be packed in accordance with Dwg. F 8837839.

5.2 Marking. Marking shall be in accordance with Dwg. F 8837839.

5.3 Shipping. When projectiles from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit. LAT round (2 HE + 6 Test, Non-Haz) may be palletized together. The projectiles are to be identified and serialized to provide traceability to Lot No., month, and year of manufacture.

6. NOTES

6.1 Intended use. Not applicable.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, ARDC, ATTN: AMSMC-QAT-I, Dover, New Jersey 07801.

6.4 Ballistic test items. Ballistic test item specified herein are described below:

a. Fully tactical HE items:

Projectile M718A1, HE, P/N 11786215
Projectile M741A1, HE, P/N 11786240

b. Projectile, Non Hazardous Test

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1. Contains inert booster and inert main charge.
2. S&A contains Dets and transfer lead only.

Projectile 155MM Test, Non-Haz,M718A1 Type , P/N 11786216
Projectile 155MM Test, Non-Haz,M741A1 Type , P/N 11786241

6.5 Projectile functioning

6.5.1 Premature burst. A premature burst is considered to have occurred if the projectile functions in the weapon or the time of flight is less than 1.9 seconds.

6.5.2 Functioning at all zones, excluding excess pressure. Satisfactory functioning of the projectile when fired at any zone, except excess pressure, is defined as successful delivery of the cargo to the target area with all projectile hardware remaining intact until initiation of the nose fuze at a pre-determined time after firing, and expulsion of the entire cargo.

6.5.3 Functioning at excess pressure. Satisfactory functioning of the projectile when fired at excess pressure is defined as successful delivery of the cargo to the target area with all projectile hardware remaining intact until initiation of the projectile fuze.

6.5.4 Excess pressure. Excess pressure in the M109A1 weapon system is herein considered to be a chamber pressure of approximately 43,000 psi.

6.6 Submission of test data and data cards. In addition to the normal distribution of records, when the projectiles are procured by the US Army Armament Material Readiness Command, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commander, ARDC, ATTN: AMSMC-QAT-M, Dover, New Jersey 07801.

6.7 Drawings. Drawings listed in section 2 of this specification under the heading US Army Armament Research and Development Command (ARDC), may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal (ARRADCOM) or Picatinny Arsenal drawings Technical data originally prepared by these activities is now under the cognizance of ARDC.

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6.9 Data requirements. Deliverable data required by the specifications are cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID No.
3.8	Plan for control in use of adhesives	DI-T-5372 (Tailored)
3.9	Safety Plans	DI-T-5372 (Tailored)
4.4.2.1, 4.4.2.4, 4.4.2.5, 4.4.2.6, 4.4.2.7, 4.4.2.8, 4.4.2.10, 4.4.2.13 4.4.2.14, 4.4.2.17	Quality Inspect Defect Reports	DI-R-1721
4.4.3.3	Quality Inspection Test, Demonstration & Evaluation Report	DI-R-1724
4.4.3.6, 4.4.3.7, 4.4.3.8,	Plans for Verification of Preload forces and Torques	DI-T-5372 (Tailored)
6.7	Ammunition Data Card	DI-L-1410

6.9 Subject term (key word) listing.

Adhesive application
Air pressure test
Assembling
Ballistic function, flight detonation
Booster, explosive performance
Electronic Assembly
Excess pressure test
Flouroscopic Examination
Handling and process plan

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Loading
Magnetic target signature test
Main charge, explosive performance
Munition
Mine, AT
Packing, Projectile, High Explosive test
Projectile function
Radiographic examination
Safe and Arm Assembly
Self-destruct function
Sensor performance test
Surveillance Units.

Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1320-A861)

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